Gael Berlanga Boemare
gaelberlanga@gmail.com — https://gaelberlan.ga — https://github.com/Gael-BB

Education

University of Cambridge Cambridge, United Kingdom
Integrated Master of Engineering - MEng, Engineering Tripos Sep. 2022 – Jul. 2026
Master Research Project: Bayesian inference for non-linear state-space models.
Bachelor grade: First Class Honours.

British School of Brussels Brussels, Belgium
A-levels: A* A* A* in Mathematics, Physics, Further Mathematics Sep. 2020 – Jul. 2022
Awards:

- European Statistics Competition 2020. 1st place, team of 3, 17,000 students, 16 countries.
- Belgian Physics Olympiad 2021. 3rd place.

Technical Skills and Projects

Algotrade Hackathon — Negative Eevee:

- Won 1st place and 5,000 EUR prize money.
- Built a high-speed arbitrage algorithm in Rust to detect and exploit pricing inefficiencies in real time.

IMC Trading Competition Prosperity 2:

- Placed top 1% out of 10,000 teams in a team of 4.
- Produced Algorithms to trade virtual products and options in a simulated market.
- Developed Strategies using the Black-Scholes equation, moving averages, and other indicators.
- Solved probability and game theory brain teasers.

Poker bot using Counterfactual Regret Minimization:

- Produced a bot that can play Kuhn and Texas Hold 'em poker variants.
- Trained using counterfactual regret minimization with abstraction of the game.

Experience

Quant Trading Intern, Da Vinci Derivatives Jul. 2025 - Aug. 2025

- Researched relationships between crypto asset baskets.
- Designed a mid-frequency trading strategy on crypto assets using Principal Component Analysis.
- Traded volatility gaps between Nvidia and QQQ options.

Data Science Intern, Galeio Jul. 2024 - Aug. 2024

- Created vector databases by utilizing Meta's DinoV2 model to generate embeddings from small patches of large satellite images.
- Achieved one-shot object detection by performing searches through vector databases using cosine similarity.
- Developed data annotation software by selecting zero-shot detected objects with Meta's Segment Anything Model, doubling speed.

Data Science Intern, Desbrest Institute of Epidemiology and Public Health Dec. 2023 - Jan. 2024

- Developed algorithms for automated pulmonary health analysis from exhaled gases using R.
- Extracted meaningful data from curves of exhaled gases by fitting non-linear models.
- Performed principal component analyses followed by clustering algorithms to classify patients.
- Conducted a study linking coastal weather to stroke cases using R.
- Utilised multiple regression techniques, notably Lasso, to extract meaningful variables.

Engineering Intern, AGRO – OLEUM INGENIERIA S.L. Jul. 2023 - Aug. 2023

- Collaborated with a team of engineers to aid in the production and development of warehouses.
- Designed and edited plans using AutoCAD from data extracted from modelling software.

Languages: French, Spanish, and English. All native proficiency.